

ELE2038 Control Coursework

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1 Introduction

This report looks at designing a PID controller for an overhead crane. The crane has a carriage of mass m_1 that moves along a horizontal rail, with a load of mass m_2 hanging from a cable of length L . The aim is to move the hook to a position x_B^{ref} chosen by the operator, while keeping the swing angle θ as small as possible.

The crane equations are nonlinear, so the first thing to do is linearise them around an equilibrium point. From there a transfer function is worked out and a PID controller is designed and tuned. The controller is then tested on the full nonlinear system in simulation, and its robustness to different loads, bad parameter estimates, and time delays is checked.

All the Python code for the simulations is available at [this repository](#).

2 System Description

2.1 Crane Dynamics

The crane carriage of mass m_1 moves horizontally at position x_A . A load of mass m_2 hangs from a cable of length L at angle θ from vertical. A horizontal force F drives the carriage. There is also linear damping $-b_x \dot{x}_A$ on the carriage and rotational friction $-b_\theta \dot{\theta}$ on the cable. The equations of motion are

$$m_{\text{tot}} \ddot{x}_A + b_x \dot{x}_A + m_2 L (\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) = F, \quad (2.1)$$

$$m_2 L \cos \theta \ddot{x}_A + m_2 L^2 \ddot{\theta} + b_\theta \dot{\theta} + m_2 g L \sin \theta = 0, \quad (2.2)$$

where $m_{\text{tot}} = m_1 + m_2$. The force F comes from an electric motor driven by a voltage $v(t) \in [-1, 1]$ V. The voltage to force transfer function is

$$G_a(s) = \frac{F(s)}{V(s)} = \frac{K_a}{(\tau_1 s + 1)(\tau_2 s + 1)}. \quad (2.3)$$

2.2 System Parameters

The parameters are $m_1 = 1033$ kg, $m_2 \in [100, 1000]$ kg, $L = 20$ m, $b_x = 820$ N s/m, $b_\theta = 2795$ N m s, $g = 9.81$ m/s², $K_a = 5200$ N/V, $\tau_1 = 270$ ms, and $\tau_2 = 10$ ms. The rail is 60 m long.

3 System Linearisation

3.1 Equilibrium Point

Since the crane equations are nonlinear, they need to be linearised before a PID controller can be designed. The first step is finding an equilibrium point, which is a state where the system just sits still. Setting all velocities and accelerations to zero in Equations (2.1) and (2.2) gives $F = 0$ from the first equation, and

$$m_2 g L \sin \theta = 0, \quad (3.1)$$

from the second, which means $\theta^* = 0$. So the equilibrium is

$$x_A^* = x_A^{\text{ref}}, \quad \theta^* = 0, \quad \dot{x}_A^* = 0, \quad \dot{\theta}^* = 0, \quad F^* = 0, \quad (3.2)$$

which makes physical sense — the cable hangs straight down when the crane is at rest.

3.2 Small Perturbation Analysis

Small deviations away from the equilibrium are introduced as

$$x_A = x_A^{\text{ref}} + \tilde{x}_A, \quad \theta = \tilde{\theta}, \quad F = \tilde{F}, \quad (3.3)$$

where the tilde denotes a small deviation. For small $\tilde{\theta}$, the standard small-angle approximations from Taylor's theorem are used,

$$\sin \tilde{\theta} \approx \tilde{\theta}, \quad \cos \tilde{\theta} \approx 1, \quad \dot{\theta}^2 \approx 0. \quad (3.4)$$

3.3 Linearised Equations of Motion

Substituting into Equation (2.1) and dropping any second-order terms gives

$$m_{\text{tot}} \ddot{\tilde{x}}_A + b_x \dot{\tilde{x}}_A + m_2 L \ddot{\tilde{\theta}} = \tilde{F}, \quad (3.5)$$

and doing the same for Equation (2.2) gives

$$m_2 L \ddot{\tilde{x}}_A + m_2 L^2 \ddot{\tilde{\theta}} + b_\theta \dot{\tilde{\theta}} + m_2 g L \tilde{\theta} = 0. \quad (3.6)$$

These are the linearised equations that will be used for the rest of the design.

4 Transfer Function Derivation

4.1 Laplace Transform

Taking the Laplace transform of Equations (3.5) and (3.6) with zero initial conditions gives

$$m_{\text{tot}}s^2\tilde{X}_A + b_x s\tilde{X}_A + m_2 L s^2\tilde{\Theta} = \tilde{F}, \quad (4.1)$$

$$m_2 L s^2\tilde{X}_A + m_2 L^2 s^2\tilde{\Theta} + b_\theta s\tilde{\Theta} + m_2 g L\tilde{\Theta} = 0. \quad (4.2)$$

4.2 Solving for $\tilde{\Theta}$

Rearranging Equation (4.2) to get $\tilde{X}_A$ on its own,

$$\tilde{X}_A = \frac{-(m_2 L^2 s^2 + b_\theta s + m_2 g L)}{m_2 L s^2} \tilde{\Theta}. \quad (4.3)$$

Substituting this into Equation (4.1) and multiplying through by $m_2 L s^2$,

$$\begin{aligned} & -m_{\text{tot}}s^2(m_2 L^2 s^2 + b_\theta s + m_2 g L)\tilde{\Theta} \\ & -b_x s(m_2 L^2 s^2 + b_\theta s + m_2 g L)\tilde{\Theta} + m_2^2 L^2 s^4\tilde{\Theta} = m_2 L s^2 \tilde{F}. \end{aligned} \quad (4.4)$$

Grouping by powers of s and using the fact that $m_2^2 L^2 - m_{\text{tot}}m_2 L^2 = -m_1 m_2 L^2$,

$$\begin{aligned} & \tilde{\Theta}[-m_1 m_2 L^2 s^4 - (m_{\text{tot}}b_\theta + b_x m_2 L^2)s^3 \\ & - (m_{\text{tot}}m_2 g L + b_x b_\theta)s^2 - b_x m_2 g L s] = m_2 L s^2 \tilde{F}. \end{aligned} \quad (4.5)$$

Factoring out $-s$ and solving for $\tilde{\Theta}$,

$$\tilde{\Theta}(s) = \frac{-m_2 L s}{D(s)} \tilde{F}(s), \quad (4.6)$$

where

$$D(s) = a_3 s^3 + a_2 s^2 + a_1 s + a_0, \quad (4.7)$$

with $a_3 = m_1 m_2 L^2$, $a_2 = b_x m_2 L^2 + b_\theta m_{\text{tot}}$, $a_1 = m_2 g L m_{\text{tot}} + b_x b_\theta$, and $a_0 = m_2 b_x g L$.

4.3 Solving for $\tilde{X}_A$

Putting Equation (4.6) back into Equation (4.3),

$$\tilde{X}_A(s) = \frac{m_2 L^2 s^2 + b_\theta s + m_2 g L}{s D(s)} \tilde{F}(s). \quad (4.8)$$

4.4 Hook Position Transfer Function

The hook position is $x_B = x_A + L \sin \theta$. Linearising gives $\tilde{x}_B = \tilde{x}_A + L\tilde{\theta}$, or in Laplace form,

$$\tilde{X}_B(s) = \tilde{X}_A(s) + L\tilde{\Theta}(s). \quad (4.9)$$

Substituting in Equations (4.6) and (4.8),

$$\begin{aligned} \tilde{X}_B(s) &= \frac{m_2 L^2 s^2 + b_\theta s + m_2 g L}{s D(s)} \tilde{F} + L \cdot \frac{-m_2 L s}{D(s)} \tilde{F} \\ &= \frac{(m_2 L^2 s^2 + b_\theta s + m_2 g L) - m_2 L^2 s^2}{s D(s)} \tilde{F}, \end{aligned} \quad (4.10)$$

which simplifies down to

$$\frac{\tilde{X}_B(s)}{\tilde{F}(s)} = \frac{b_\theta s + m_2 g L}{s(a_3 s^3 + a_2 s^2 + a_1 s + a_0)}, \quad (4.11)$$

which matches the expected form from the problem statement.

5 Block Diagram

5.1 Closed-Loop Structure

Figure 1 shows how all the components connect together. The operator sets a desired hook position x_B^{ref} , which is compared to the measured output to get the error $e(s)$. The PID controller $C(s)$ uses this error to produce a voltage $v(t)$, which the actuator $G_a(s)$ converts to a force F . This force drives the crane plant $G_p(s)$ to move the hook to $\tilde{X}_B$. Wind is modelled as a disturbance $D(s)$ entering at the plant input. The sensor $G_s(s)$ models the small delay from computing $x_B = x_A + L \sin \theta$ using the on-board sensors.

5.2 Transfer Functions

The full forward path from voltage to hook position is

$$G(s) = G_a(s) G_p(s) = \frac{K_a}{(\tau_1 s + 1)(\tau_2 s + 1)} \cdot \frac{b_\theta s + m_2 g L}{s(a_3 s^3 + a_2 s^2 + a_1 s + a_0)} \quad (5.1)$$

The closed-loop transfer function is

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)G_s(s)}, \quad (5.2)$$

and the transfer function from disturbance to output is

$$T_d(s) = \frac{G_p(s)}{1 + C(s)G(s)G_s(s)}. \quad (5.3)$$

6 PID Controller Design

6.1 Controller Structure

The PID controller takes the form

$$C(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s}. \quad (6.1)$$

The controller was designed using the linearised model with a nominal load of $m_2 = 500$ kg.

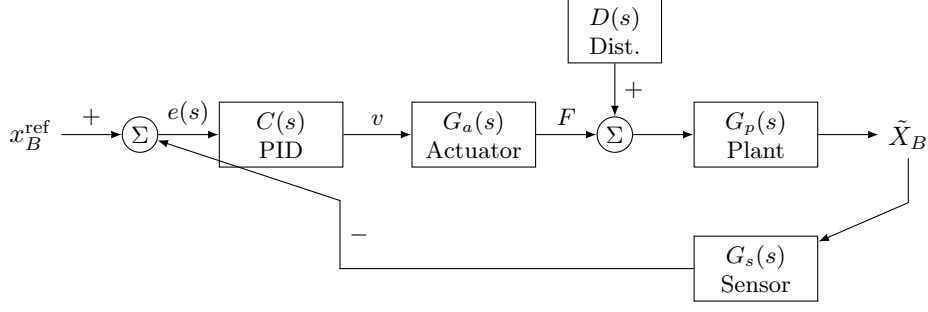


Figure 1: Block diagram of the closed-loop crane control system. The PID controller $C(s)$ drives the actuator $G_a(s)$, which applies a force to the plant $G_p(s)$. Wind disturbance $D(s)$ enters at the plant input and the sensor $G_s(s)$ models the measurement delay.

6.2 Steady-State Error

Using the Final Value Theorem, for a unit step reference the steady-state error is

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s}{1 + C(s)G(s)G_s(s)} \cdot \frac{1}{s} = 0, \quad (6.2)$$

because the plant already has one integrator and the K_i/s term in the PID adds a second one. Two integrators in the open loop means zero steady-state error for any step input, regardless of the gain values chosen.

6.3 Controller Tuning

The gains were found by running a grid search over different combinations of K_p , K_i , and K_d , keeping only results where the gain margin was above 6 dB and the phase margin above 45° . The best set of gains found was

$$K_p = 0.009, \quad K_i = 0.00012, \quad K_d = 0.008. \quad (6.3)$$

The Bode plot of the open-loop system with this controller is shown in Figure 2. The gain margin is 12.85 dB at 0.7365 rad/s and the phase margin is 72.59° at 0.0580 rad/s, both comfortably above the targets.

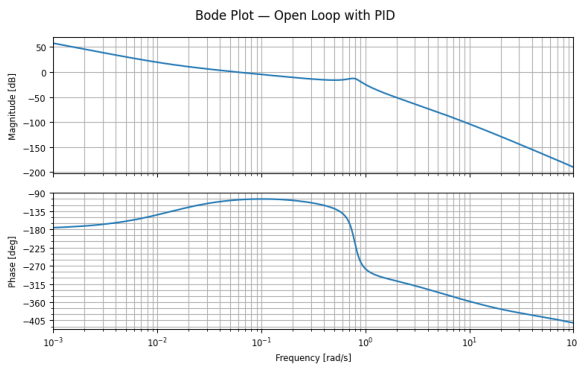


Figure 2: Bode plot of the open-loop system $C(s)G(s)G_s(s)$ with the tuned PID controller. Gain margin is 12.85 dB and phase margin is 72.59° .

6.4 Stability Proof

To confirm stability, the closed-loop poles were computed and are shown in Table 1. Every pole has a negative real part, which means they are all in the left half plane and the system is stable.

Table 1: Closed-loop poles with the tuned PID controller.

Pole	Stable?
-100.000	Yes
-20.000	Yes
-3.702	Yes
-0.604	Yes
$-0.076 \pm 0.769j$	Yes
-0.038	Yes
-0.021	Yes

7 Simulation Results

7.1 Linearised Model Response

Figure 3 shows the step response of the closed-loop system using the linearised model. The hook position tracks the reference with a rise time of 24.05 s, a settling time of 194.00 s, and an overshoot of 13.79%. The carriage position x_A behaves similarly, with a rise time of 23.81 s and a settling time of 193.50 s, which makes sense since $\theta \rightarrow 0$ at steady state so x_A and x_B end up at the same position.

7.2 Disturbance Rejection

Figure 4 shows what happens when a step disturbance hits the plant input. The hook only moves by 0.016 m at most and comes straight back to zero, which shows the integral term is doing its job.

7.3 Nonlinear Simulation

The controller was also tested on the full nonlinear system from Equations (2.1) and (2.2), using Euler

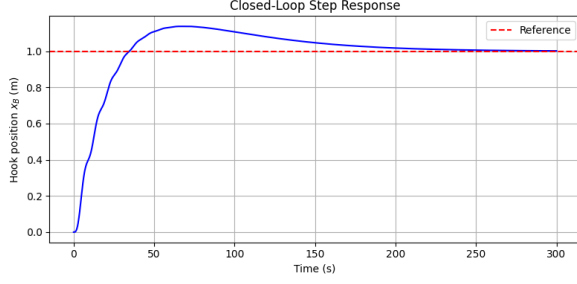


Figure 3: Step response of the closed-loop linearised system. The hook reaches the reference with 13.79% overshoot and settles in 194 s.

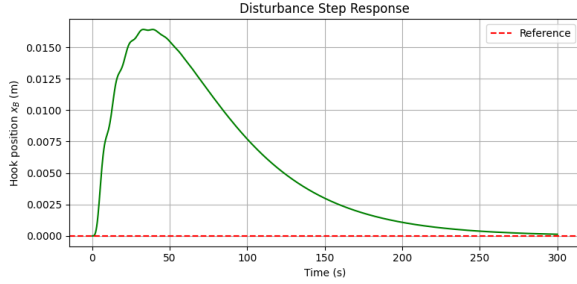


Figure 4: Response to a unit step disturbance at the plant input. Maximum deviation is 0.016 m and the output returns to zero.

integration with a step size of $\Delta t = 0.0375$ s. Figure 5 shows the results.

The hook settles at 1.0022 m, which is very close to the 1 m reference. The carriage ends up at the same spot since θ goes to zero. The maximum swing angle was only 0.179° , which is tiny and perfectly safe. The nonlinear response looks almost identical to the linearised one, which confirms the linearisation was a good approximation.

8 Robustness Analysis

8.1 Load Variation

The controller was designed for $m_2 = 500$ kg but needs to work for loads anywhere between 100 and 1000 kg. Table 2 shows the margins and performance across that range. The system stays stable with good margins throughout, and the settling time and overshoot barely change, which is a good sign.

Table 2: Stability margins for different load masses.

m_2 (kg)	GM (dB)	PM ($^\circ$)	OS (%)
100	7.01	74.23	14.35
500	12.85	72.59	13.79
1000	19.02	70.56	14.21

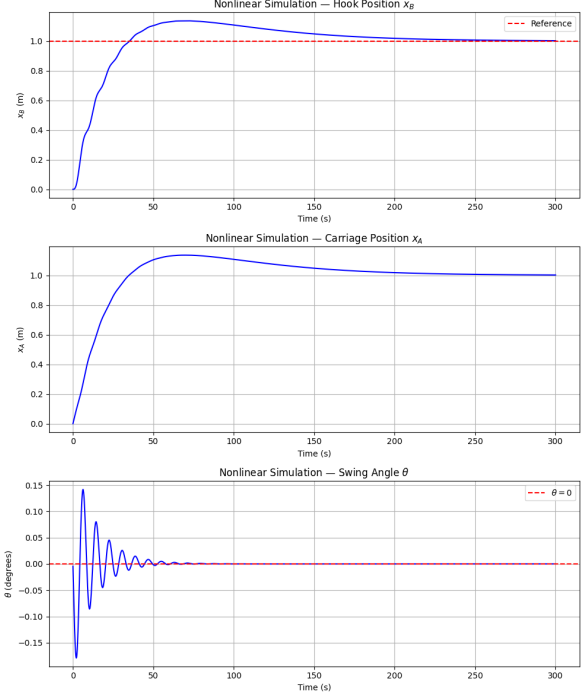


Figure 5: Nonlinear simulation with $m_2 = 500$ kg showing hook position x_B , carriage position x_A , and swing angle θ . Maximum swing is 0.179° and the hook settles to within 0.003 m of the reference.

8.2 Parameter Misestimation

Each parameter was varied independently by factors of 0.5 up to 2.0 to see how much the system could handle before going unstable. It stayed stable in every case tested. The two parameters that had the biggest effect were K_a and L — doubling either one brought the gain margin down to around 7 dB, which is still just above the 6 dB limit. The time constants τ_1 and τ_2 had very little effect, and τ_2 barely changed the margins at all. The damping terms b_x and b_θ were also not a big issue.

8.3 Time Delays

Sensor and actuator delays were modelled using a second-order Padé approximation. The system stayed stable all the way up to a 5 s delay, which is way more than any real crane sensor would have. Figure 6 shows the step response gets a bit slower with larger delays but the system never goes unstable.

8.4 Summary

Overall the controller holds up well across all the tests. It works for the full load range, handles parameters being off by up to a factor of two, and deals with delays of at least 5 s. The one parameter to watch is K_a since doubling it gets close to

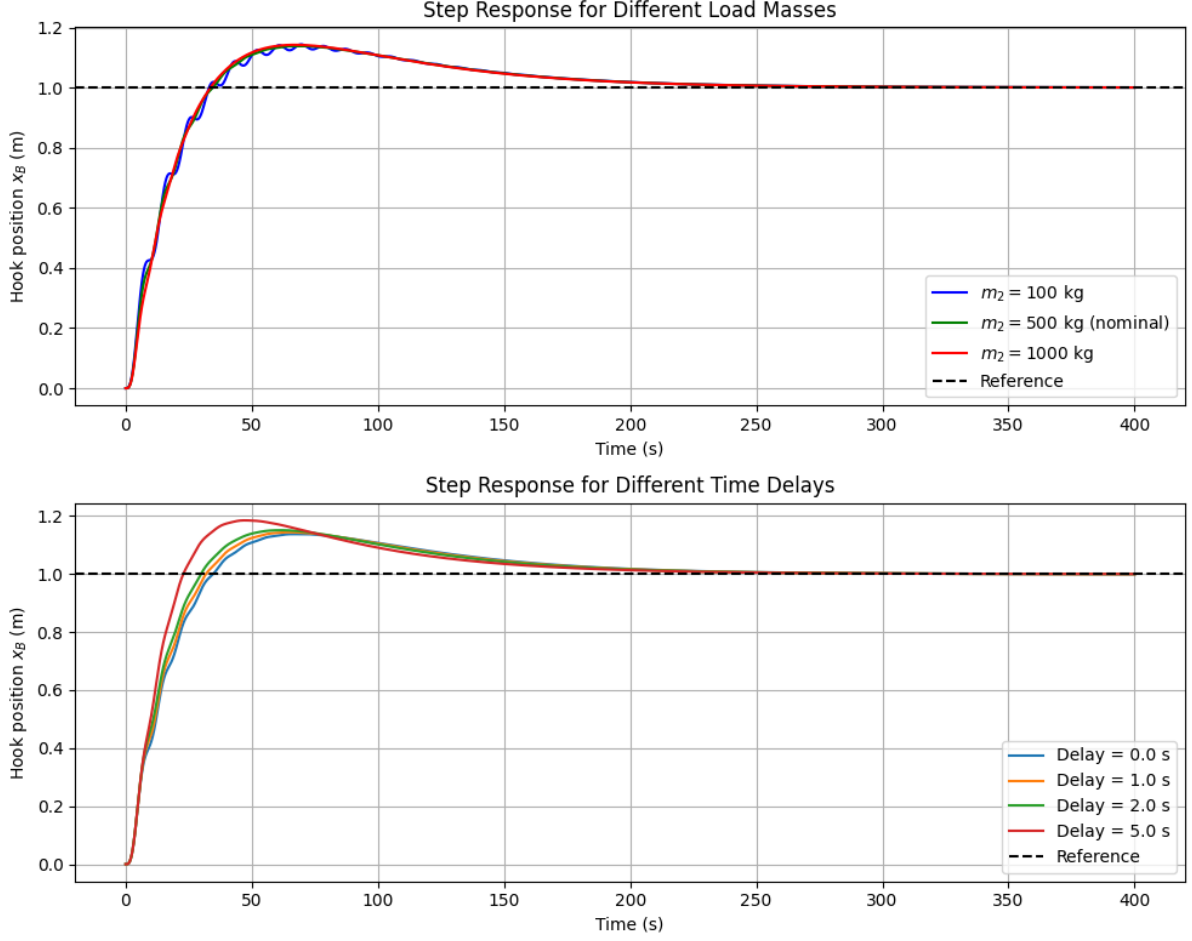


Figure 6: Robustness results. Top: step responses for $m_2 \in \{100, 500, 1000\}$ kg showing consistent performance. Bottom: step responses for delays of 0, 1, 2, and 5 s showing the system stays stable throughout.

the stability limit, so it would be worth making sure that is measured accurately on the real crane.

9 Conclusion

A PID controller was designed for an overhead crane using a model linearised around the equilibrium $\theta^* = 0$. The final gains are $K_p = 0.009$, $K_i = 0.00012$, and $K_d = 0.008$, giving a gain margin of 12.85 dB and a phase margin of 72.59° . The controller gives zero steady-state error for step inputs and all closed-loop poles are in the left half plane confirming stability.

Testing on the full nonlinear model showed the hook settling to within 0.003 m of the target with a maximum swing angle of just 0.179° . The controller works well across loads from 100 to 1000 kg, handles parameter errors of up to a factor of two, and stays stable for sensor delays up to 5 s. The actuator gain K_a is the most sensitive parameter and should be calibrated carefully in practice.